Possible Answers :

1

Question Number : 25 Question Id : 695278325 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $y = y(x)$ be the solution of the differential equation $x\sin\left(\frac{y}{x}\right)dy = \left(y\sin\left(\frac{y}{x}\right) - x\right)dx$, $y(1) = \frac{\pi}{2}$ and let $\alpha = \cos\left(\frac{y\left(e^{12}\right)}{e^{12}}\right)$. Then the number of integral value of p, for which the equation $x^2 + y^2 - 2px + 2py + \alpha + 2 = 0$ represents a circle of radius $r \leq 6$, is_____.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

Physics Section A

Section Id :	69527827
Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69527827
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 26 Question Id : 695278326 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In a Vernier calipers, when both jaws touch each other, zero of the Vernier scale is shifted to the right of zero of the main scale and 7th Vernier division coincides with a main scale reading. If the value of 1 main scale division is 1 mm and there are 10 Vernier scale divisions, then the Vernier caliper has

Options :

6952781106. 0.07 cm negative zero error

6952781107. 0.7 cm negative zero error

6952781108. 0.07 cm positive zero error

6952781109. 0.7 cm positive zero error

Question Number : 27 Question Id : 695278327 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

L, C and R represents physical quantities inductance, capacitance and resistance respectively. The dimensional formula $M L^2 T^{-4} A^{-2}$ corresponds to _____.

Options :

6952781110. $\frac{R}{\sqrt{LC}}$

6952781111. $\frac{R}{LC}$

6952781112. $\frac{C}{\sqrt{LR}}$

6952781113. $\frac{1}{R}\sqrt{\frac{L}{C}}$

Question Number : 28 Question Id : 695278328 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

When one moves from a point 16 km below the earth's surface to a point 16 km above the earth's surface. The change in g is approximately α %. The value of α is _____.

(Take radius of the earth = 6400 km.)

Options :

6952781114. 0.12

6952781115. 0.25

6952781116. 0.50

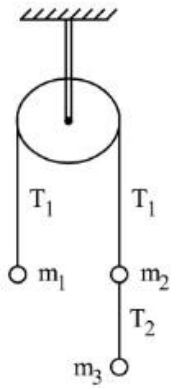
6952781117. 0.75

Question Number : 29 Question Id : 695278329 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Three masses $m_1 = 4$ kg, $m_2 = 4$ kg and $m_3 = 6$ kg are suspended from a fixed smooth frictionless pully as shown in the figure below. The value of T_1/T_2 is

_____

(take $g = 10$ m/s^2)



Options :

6952781118. 5/3

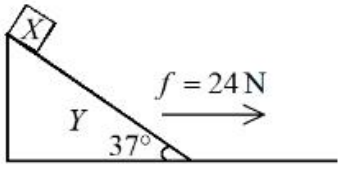
6952781119. 2/3

6952781120. $3/5$

6952781121. $2/5$

Question Number : 30 Question Id : 695278330 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A wedge Y with mass of 10 kg and all frictionless surfaces and the inclined surface making 37° with horizontal. A block X with mass 2 kg is placed at the highest point of the wedge as shown in figure is at rest. At $t = 0$ wedge (Y) is pulled toward right with constant force (f) of 24 N. Taking the block X at rest at $t = 0$, the time taken by it to slide down 8.8 m on the slope, while Y is on the move, is_____ s.
(take tan (37°) = 3/4 and g = 10 m/s^2)



Options :

6952781122. 2

6952781123. 4

6952781124. $\sqrt{2}$

6952781125. $2\sqrt{2}$

Question Number : 31 Question Id : 695278331 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The Young's modulus of steel wire of radius r and length L is Y.

If the radius r and length L of the wire are doubled then the value of Y

Options :

6952781126. increases by two times

6952781127. reduces by half

6952781128. remains unchanged

6952781129. becomes one fourth

Question Number : 32 Question Id : 695278332 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements: one is labelled as **Assertion A** and the other is labelled as **Reason R**

Statement I: Change in internal energy of a system containing n mole of ideal gas can be written as $\Delta U = n\, C_v \left(T_f - T_i\right) = \frac{nR}{\gamma - 1}\left(T_f - T_i\right)$, where $\gamma = \frac{C_p}{C_v}$, T_i = initial temperature, T_f = final temperature.

Statement II: Relation between degree of freedom f and γ $(= C_p/C_v)$ is $\left(\gamma = 1 + \frac{2}{f}\right)$

Choose the ***correct*** answer from the options given below

Options :

6952781130. Both **A** and **R** are true and **R** is the correct explanation of **A**

6952781131. Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**

6952781132. **A** is true but **R** is false

6952781133. **A** is false but **R** is true

Question Number : 33 Question Id : 695278333 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the following statements:

A. Zeroth law of thermodynamics gives concept of temperature
B. First law of thermodynamics gives concept of internal energy
C. In isothermal expansion of ideal gas, $\Delta Q \neq \Delta W$
D. Product of intensive and extensive variables is extensive
E. The ratio of any extensive variable to mass will be an extensive variable

Choose the correct combination of statements from the options given below:

Options :

6952781134. C, D and E Only

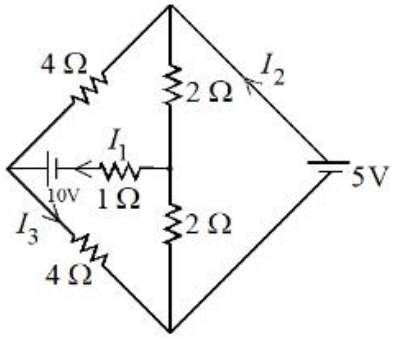
6952781135. A, B and C Only

6952781136. A, B and D Only

6952781137. B, C and D Only

Question Number : 34 Question Id : 695278334 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Refer to the figure given below. The values of I_1, I_2 and I_3 are _____.



Options :

6952781138. $I_1 = 2.5$ A, $I_2 = 1.875$ A, $I_3 = 1.875$ A

6952781139. $I_1 = 1.875$ A, $I_2 = 2.5$ A, $I_3 = 1.875$ A

6952781140. $I_1 = 1.875$ A, $I_2 = 1.875$ A, $I_3 = 2.5$ A

6952781141. $I_1 = 2.5$ A, $I_2 = 2.5$ A, $I_3 = 1.875$ A

Question Number : 35 Question Id : 695278335 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

An electron of mass m is moving in an electric field $\vec{E} = -2E_o\hat{i}$ (E_o = constant > 0), with an initial velocity $\vec{V} = v_o\hat{i}$ (v_o = constant > 0). If $\lambda_o = \frac{h}{4mv_o}$, its de Broglie wavelength at time t is _____.
(e = charge of electron)

Options :

6952781142. $\frac{4\lambda_o}{\left[1-\frac{E_o e}{2m}\frac{t}{v_o}\right]}$

6952781143. $\frac{4\lambda_o}{\left[1+\frac{E_o e}{2m}\frac{t}{v_o}\right]}$

6952781144. $\frac{4\lambda_o}{\left[1+\frac{2E_o e}{m}\frac{t}{v_o}\right]}$

6952781145. $\frac{4\lambda_o}{\left[1-\frac{2E_o e}{m}\frac{t}{v_o}\right]}$

Question Number : 36 Question Id : 695278336 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In the hydrogen atom, the electron makes a transition from the higher orbit (i) to a lower orbit (f). The ratio of the radius of the orbits in given by $r_i : r_f = 16 : 4$. The wavelength of photon emitted due to this transition is _____nm.
(Given Rydberg constant = 1.0973×10^7 /m)

Options :

6952781146. 121

6952781147. 242

6952781148. 486

6952781149. 974

Question Number : 37 Question Id : 695278337 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A displacement current of 4.0 A can be set up in the space between two parallel plates of 6 μF capacitor. The rate of change of potential difference across the plates of the capacitor is nearly $\alpha \times 10^6$ V/s. The value of α is _____.

Options :

6952781150. 0.58

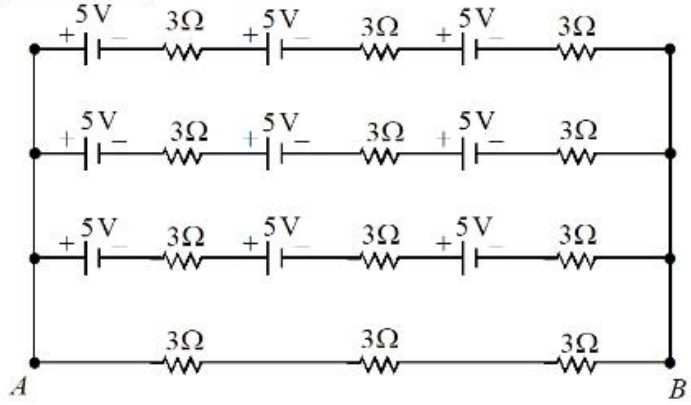
6952781151. 0.67

6952781152. 0.82

6952781153. 0.75

Question Number : 38 Question Id : 695278338 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Refer to the figure given below, current between terminals A and B is _____A.



Options :

6952781154. 12.5

6952781155. 1.25

6952781156. 7.5

6952781157. 5

Question Number : 39 Question Id : 695278339 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In Young's double slit experiment, the fringe width of the interference pattern produced on the screen is 2.4 μm. If the experiment is carried out in another medium having refractive index 1.2, the fringe width will be ____ μm.

Options :

6952781158. 1.2

6952781159. 2

6952781160. 2.4

6952781161. 2.88

Question Number : 40 Question Id : 695278340 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A ray of light passing through an equilateral prism is having velocity 2.12×10^8 m/s in the prism material, then the minimum angle of deviation is _____ degrees.

Options :

6952781162. 45

6952781163. 30

6952781164. 28

6952781165. 58

Question Number : 41 Question Id : 695278341 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Light source having wavelength 331 nm is used to generate photo-electrons whose stopping potential is 0.2 V. The work function of the used metal in the experiment is $\alpha \times 10^{-19}$ J. The value of α is _____.
($h = 6.62 \times 10^{-34}$ J s, $e = 1.6 \times 10^{-19}$ C and $c = 3 \times 10^8$ m/s)

Options :

6952781166. 3.68

6952781167. 4.68

6952781168. 5.68

6952781169. 2.68

Question Number : 42 Question Id : 695278342 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A compound microscope is designed with two symmetric biconvex lenses. The objective lens is cut vertically, creating two identical plano-convex lenses. One of them is used in place of original objective lens. To retain same magnification keeping the object distance unchanged, the tube length has to be

Options :

6952781170. increased two times

6952781171. increased $\frac{3}{2}$ times

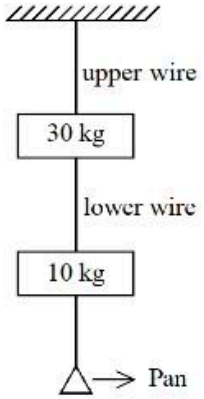
6952781172. decreased two times

6952781173. decreased $\frac{3}{2}$ times

Question Number : 43 Question Id : 695278343 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two wires as shown in the figure below, made of steel and have breaking stress of 12×10^8 N/m^2. Area of cross-section of upper wire is 0.008 cm^2 and of lower wire is 0.004 cm^2. The maximum mass that can be added to pan without breaking any wire is _____ kg.

(take $g = 10$ m/s^2)



Options :

6952781174. 56

6952781175. 38

6952781176. 96

6952781177. 5.6

Question Number : 44 Question Id : 695278344 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

An a.c. source of angular frequency ω is connected across a resistor R and a capacitor C in series. The current is observed as I. Now the frequency of the source is changed to $\omega/4$, (keeping the voltage unchanged) the current is found to be $I/3$. The ratio of resistance to reactance at frequency ω is

Options :

6952781178. $\sqrt{\frac{6}{7}}$

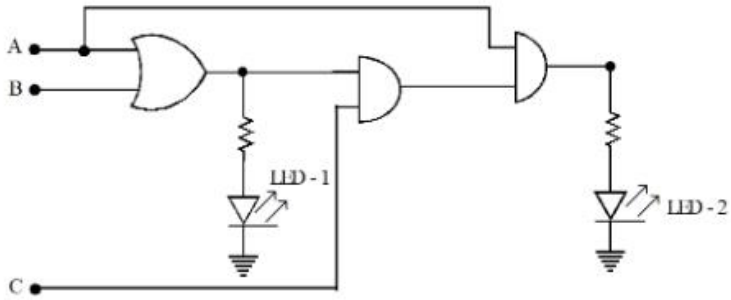
6952781179. $\sqrt{\frac{3}{5}}$

6952781180. $\sqrt{\frac{7}{8}}$

6952781181. $\sqrt{\frac{3}{4}}$

Question Number : 45 Question Id : 695278345 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

For the given logic circuit, which of the following inputs combination will make both LED−1 and LED−2 to glow?



Options :

6952781182. $A = 0, B = 1, C = 1$

6952781183. $A = 1, B = 0, C = 0$

6952781184. $A = 1, B = 0, C = 1$

6952781185. $A = 1, B = 1, C = 0$

Physics Section B

Section Id :	69527828
Section Number :	4
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	69527828
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 46 Question Id : 695278346 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A cube has side length 5 cm and modulus of rigidity 10^5 N/m^2. The displacement produced by a force of 10 N in the upper face of cube is _____ mm.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Question Number : 47 Question Id : 695278347 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

From 18 m height above the ground a ball is dropped from rest . The height above the ground at which the magnitude of velocity equal to the magnitude of acceleration (in the same set of units) due to gravity is ____ m.

(Take $g = 10\ m/s^2$ and neglect the air resistance)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Question Number : 48 Question Id : 695278348 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A transverse wave on a string is described by $y = 3\sin(36t + 0.018x + \pi/4)$. where x, y are in cm and t in seconds. The least distance between the two successive crests in the wave is ____ cm. (Nearest integer)

$(\pi = 3.14)$

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Question Number : 49 Question Id : 695278349 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The charged particle moving in a uniform magnetic field of $\left(3\hat{i} + 2\hat{j}\right)$T has an acceleration $\left(4\hat{i} - \frac{x}{2}\hat{j}\right) m/s^2$. The value of x is

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

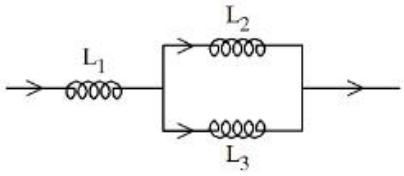
Possible Answers :

1

Question Number : 50 Question Id : 695278350 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

In the given circuit below inductance values of L_1, L_2 and L_3 are same. The magnetic energy stored in the entire circuit is (U_t) and that stored in the L_2 inductor is (U_l). U_t / U_l is _____.

(Ignore the mutual inductance if any)



Response Type : Numeric

Evaluation Required For SA : Yes